Al Mualla Al Maawali

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Final-year Electrical & Computer Engineering student (Computer Systems & Networks) at Sultan Qaboos University. Co-author of a satellite AI system now in orbit; lead engineer on an award-winning autonomous robot.

Selected Projects

- **SpaceBorne AI — satellite land-use algorithm (deployed to space).**
  Developed the full classification stage end-to-end and led subsystem integration. (Ministry of Housing). Achieved 90% classification accuracy. Deployed via the Oman Lens Rideshare Program in collaboration with STAR.VISION (China).
- **Task-Agnostic Modular Robotic Apparatus — autonomous indoor service robot (final-year project).**
  Delivered ±10 cm navigation accuracy at a ∼73% cost reduction versus the closest commercial alternatives. Won the Engineering Gathering Prize at SQU's 16th Engineering Gathering; funded by SQU Undergraduate Research Grant. Responsible for hardware end-to-end.
- **NetScope — handheld Wi-Fi monitoring device.**
  ESP32-based embedded tool with a custom touchscreen UI; built end-to-end during an international engineering internship in Turkey.
- **Quantum — autonomous differential-drive robot.**
  ESP32-S3 platform with time-of-flight sensing and closed-loop motor control.
- **FPGA PS/2 Keyboard Decoder.**
  Verilog implementation on an Altera FPGA; ∼15% lower latency than an equivalent software handler.

Experience

- **Student Researcher** — Oman Lens Rideshare Program
  Co-developed the SpaceBorne AI system in collaboration with STAR.VISION (China); delivered the complete classification stage.
- **Engineering Intern** — Ege University, İzmir, Turkey
  Two-month technical internship (IAESTE programme); designed and built NetScope, a handheld Wi-Fi monitoring device, as the internship deliverable.
- **Research Assistant** — Sultan Qaboos University
  Industry-partnered research with a leading Omani company; work covered machine vision, 3D reconstruction, and subsystem integration.

Education

- **Sultan Qaboos University** — Muscat, Oman
  B.Eng. in Electrical & Computer Engineering — Computer Systems & Networks specialisation.

Technical Skills

Languages/ Tools: C++, Verilog, MATLAB, Git, Python, ML (TensorFlow)
Hardware: Microcontrollers, FPGA (Altera), autonomous mobile robots, drone systems, Fusion 360 (CAD)
Domains: Embedded systems, ROS 2 & robotics, computer networks, digital design, machine learning, computer vision, IoT, Linux system administration

Certifications

- **Red Hat System Administration I (RH124)** — Red Hat Enterprise Linux 10
- **Cisco CCNA: Switching, Routing & Wireless Essentials** — Cisco Networking Academy
- **Cisco CCNA: Introduction to Networks** — Cisco Networking Academy

Awards & Activities

- **Engineering Gathering Prize** — SQU 16th Engineering Gathering (for the Task-Agnostic Modular Robotic Apparatus).
- **UNESCO Certificate of Appreciation** — awarded by the UNESCO Chair on Artificial Intelligence at CIRC for presenting the SpaceBorne AI research.
- **1st Place** — UAEU–SQU SDG Debate Competition.
- **Winner** — SQU Speech Competition.
- **Top 10 Finalist** (of 120+ teams) — Oman Competition for AI-Enhanced Drones; featured on Oman TV (YouTube).

International Experience

- **International Astronautical Congress (IAC)** — Sydney, Australia.
  Presented the SpaceBorne AI research at the world's largest annual space conference.
- **Engineering Internship** — Ege University, İzmir, Turkey.
  Two-month technical internship through the IAESTE programme; designed and built NetScope as the internship deliverable.
- **Education & Culture Program** — Sakarya University, Turkey.
  Selected as student speaker representing the SQU delegation.

Conference Papers & Presentations

- Almoulla T., Almaawali S.H., Al Ghafri S.H., Mughal M.R. — *SpaceBorne AI for Land Use and Land Change Applications.* Presented at the **International Astronautical Congress (IAC), Sydney, Australia.**
- *SpaceBorne AI: Two-Stage Algorithm for Land Use and Land Change Detection.* Presented at the **(AI)²E Conference**, Sultan Qaboos University, Muscat.